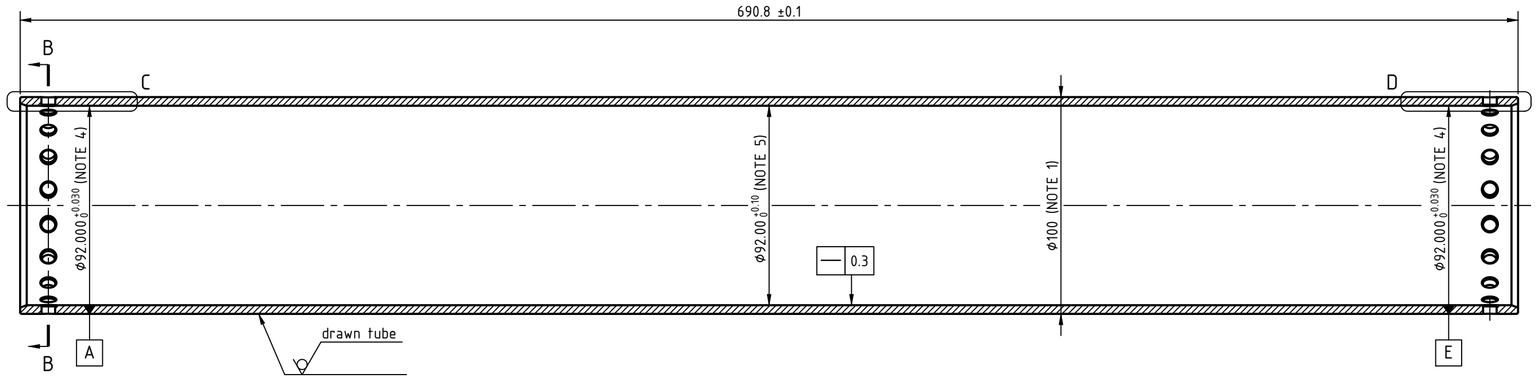
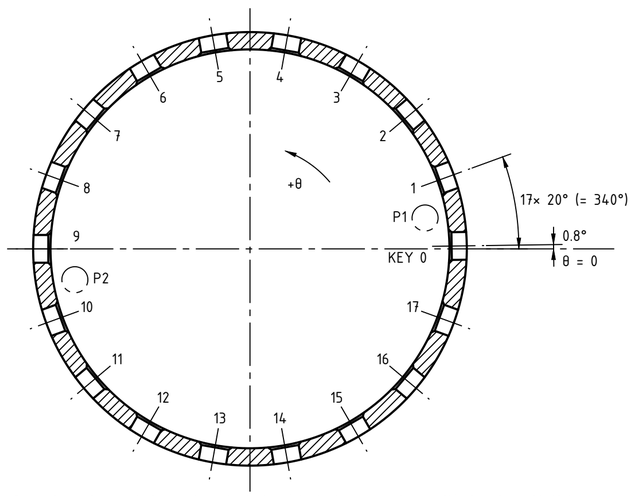
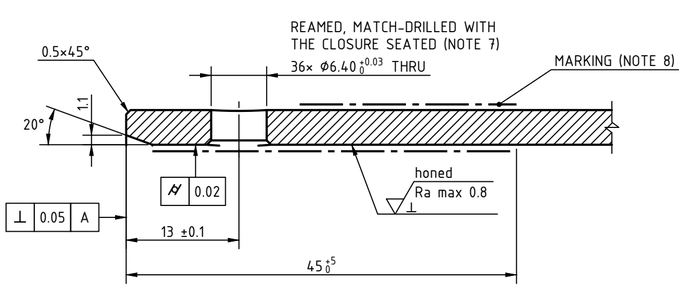


SECTION B-B (1:1)



- Seen from forward: θ counter-clockwise (clockwise seen from aft), θ = 0 = +X. Hole 0 (KEY) at 0.8°, holes 1 … 17 at 20° steps. Forward row (z 677.8, 13 ±0.1 from the fwd end face): the same pattern and numbering, its hole 0 in line with the aft hole 0 within ±2°.
- θ = 0 is set at match-drilling (op 120) from the closure's puller hole P1 at θ 10° (P2 at 190°; shown in phantom at the nozzle holder's R41, on its aft face; forward closure: R37.5, on its fwd face): hole 0 goes 9.2° clockwise of P1 seen from forward (9.2° counter-clockwise seen from aft). From then on hole 0, the clocking line and the AFT / FWD marking (note 8) carry θ = 0 on the case.

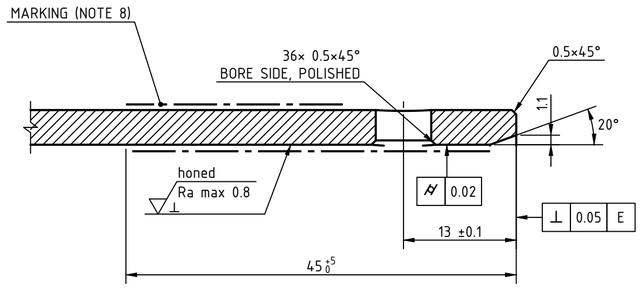
DETAIL C (2:1)



BOLT HOLES, EACH ROW

HOLE	θ ±0.1°	HOLE	θ ±0.1°
0 KEY	0.8°	9	180°
1	20°	10	200°
2	40°	11	220°
3	60°	12	240°
4	80°	13	260°
5	100°	14	280°
6	120°	15	300°
7	140°	16	320°
8	160°	17	340°

DETAIL D (2:1)



NOTES

1. Material: 6061-T6 seamless drawn tube to ASTM B210M, ϕ100 × 4.0 wall, mill certificate with heat / lot number (R1). OD and wall not machined: ϕ100 to the B210M tolerance, ISO 2768-mK does not apply to it. No heat treatment, welding, straightening or stamping.
2. Finish: none. The case is left bare (no anodising, paint or coating). Clean, oil-free, no marking ink or tape residue left on it.
3. Tube acceptance (op 40, R2, R3): wall ≥ 3.80 everywhere and bore ϕ ≤ 92.10 everywhere (two-point), the honed seal zones included (note 4 keeps them there). Reject the tube otherwise.
4. Seal zones (thick chain lines, details C, D; drawn at A = 92.000): hone each zone to the smallest diameter that cleans it up, A, and hold it to A +0.030/0, Ra 0.4 (Ra 0.8 max), circumferential lay. A = the smallest of the zone's six R4 readings, recorded per zone and marked next to it (note 8). Accept 92.000 ≤ A ≤ 92.070 (so no zone exceeds 92.10), else reject the tube. Closures finished to the A of their zone: plug land A − 0.06 (0/−0.02), O-ring groove bottom A − 5.67 (0/−0.04), i.e. 91.94 / 86.33 at A = 92.000 (SRM100-102, -103). Hone run-out blended into the as-drawn bore, no step. Datum A / E: axis of the aft / fwd honed zone.
5. Bore between the seal zones as drawn: ϕ92.00 … 92.10 (the clearance parts are sized to it), surface lines straight within 0.3. A GO plug ϕ91.70 × 150 and liner SRM100-110 must pass the full length by hand. A bore under 92.00 is honed to 92.00 … 92.10 full length.
6. Length 690.8 ±0.1 closes the axial stack (SRM100-000): 2 closure plugs, 2 insulators, 4 grains ±0.1, 5 spacers ±0.02 and the take-up ring, 3 installed. Stack ±0.5 + case ±0.1 gives a take-up gap of 2.4 … 3.6, i.e. 50 … 25 % compression of the 4.8 silicone sponge (limit 25 … 50 %). The gap is measured at assembly.
7. Bolt holes ONLY at assembly (ops 110 … 150), the mating closure seated. Each case end and its closure are a serialised pair: never interchange closures or turn the case end for end.
8. Marking (op 160), laser or dot-peen, 0.05 deep max, 3 mm characters: on the OD next to hole 0 at each end, z 20 … 45 from the end face, centred on θ = 0, reading round the tube: 'SRM100-101 S/N …' over 'AFT A 92.xxx' (aft end) or 'FWD A 92.xxx' (fwd end), A of that zone (note 4). Clocking line (op 140): scribed 0.05 deep max at θ = 0, z 0 … 8, continued across the closure flange. Temperature labels, fitted new at assembly for every firing: irreversible 177 / 204 / 232 / 260 °C strips on the OD at z 20 … 35 from the aft face, θ 90° and 270°, clear of the marking and head sweep (two more on the holder, SRM100-102); read and recorded before extraction (R10).
9. Re-use: the case is re-used after inspection (R10) after every firing. Proof test on SRM100-001 (R9) before first use and after any repair, any retired-and-replaced closure, or a temperature label reading ≥ 232 °C. The case record R1 … R10 stays with the case serial for its life, with the closure pair serials.
10. Edges ISO 13715 -0.3 / +0.3 unless stated.
11. USED ON: SRM100-000 motor, item 1 (1 off); SRM100-001 hydrostatic proof assembly, item 1.

OPERATIONS

OP	LOOSE TUBE (as supplied to assembly)	OP	AT ASSEMBLY SRM100-000, each end with its closure seated
10	Receive the tube with its mill certificate (note 1, R1). Cut and face both ends to 690.8 ±0.1; end faces square to the bore (⊥ 0.05 A / E, checked after op 30).	110	Take the AFT end (tape label) and the nozzle holder SRM100-102 finished to its A (FWD end: forward closure SRM100-103). Seat the closure in the bore, flange on the end face, O-rings not fitted. Clamp the pair.
20	Turn the 0.5×45° OD break and bore the 20°×1.1 radial O-ring lead-in at both ends (details C, D); lead-in blended into the bore, no step, Ra 0.8.	120	Clock hole 0 from the closure's puller hole P1 (θ 10°): 9.2° clockwise of P1 seen from forward, 9.2° counter-clockwise seen from aft (SECTION B-B). Drill ϕ5 (tap drill) through the case wall into the closure, depth stops per SRM100-102 / -103: hole 0 first, then holes 1 … 17.
30	Label the ends AFT and FWD (removable tape). Measure each seal zone (two-point). Hone the zone, 45 +5/0 from the end face, to the smallest diameter that cleans it up: Ra 0.4 (0.8 max), circumferential lay, cylindricity 0.02, run-out blended into the as-drawn bore. A = the smallest of its six R4 readings; accept 92.000 ≤ A ≤ 92.070, zone A +0.030/0 (note 4). Record A (R4) and write it on the tape label.	130	Case wall only, closure still in place: drill ϕ6.2 and ream ϕ6.40 +0.03/0, piloted in the ϕ5 hole; stop when the full diameter is through the case bore (the entry cone left in the closure, 0.5 deep max, lies in its first, uncounted thread). Tap the closure M6 through the case holes (per SRM100-102 / -103).
40	Inspect and record (R2 … R6): UT wall ≥ 3.80 and bore ϕ ≤ 92.10 (note 3); seal zones A +0.030/0; mid bore, straightness, GO plug and liner trial (note 5); length and ⊥.	140	Scribe the clocking line at θ = 0 across the case end and the closure flange; mark the closure per SRM100-102 note 8 / SRM100-103 note 11 (pair serial included); the case end is marked at op 160 (note 8). Remove the closure.
50	Clean (IPA), cap the ends, tag with the case serial and both A values. The tube leaves this stage with NO bolt holes and NO permanent marking; the closures are finished to the A of their end (SRM100-102, -103).	150	Break every hole on the bore side 0.5×45° and polish (the O-rings are dragged over it); deburr the OD edges. Remove all chips; re-inspect the seal zone. Repeat ops 110 … 150 at the FWD end, the forward closure turned so that its P1 lies 9.2° counter-clockwise (seen from forward) of the aft hole 0 line, transferred along the tube, ±2°.
		160	Remove the tape labels and mark the case (note 8). Record the hole baseline (R7). Proof test on SRM100-001 (R9), then take the hardness / conductivity baseline (R8).

CASE RECORD (note 9)

REC	CHECK	WHEN	WHERE / HOW	ACCEPT
R1	Mill certificate	op 10	ASTM B210M, 6061-T6, heat / lot number	per B210M
R2	Wall, ultrasonic map	op 40	8 azimuths every 50 along the full length; all round each bolt row, z 5 … 21 from each end face	wall ≥ 3.80 everywhere
R3	Bore, two-point	op 30 (pre-hone), op 40	every 50 along the length, 2 directions 90° apart	≤ 92.10 everywhere, honed zones included (R4); mid bore note 5
R4	Seal zones AFT, FWD: A and diameters	op 30, op 40	z 10, 25, 40 from the end face × 2 directions 90° apart, to 0.001: A = the smallest of these 6 readings, per zone; Ra; A also marked on the tube (note 8)	92.000 ≤ A ≤ 92.070; zone A +0.030/0 (so ≤ 92.10); Ra ≤ 0.8
R5	Mid bore	op 40	surface-line straightness; GO plug ϕ91.70 × 150 over the full length; liner SRM100-110 slide trial	note 5
R6	Length, end faces	op 40	length at 4 azimuths; each end face to its datum A / E	690.8 ±0.1; ⊥ 0.05
R7	Hole baseline, 36 holes	op 160, before the first proof; after every proof	per hole: axial and circumferential ϕ, ligament end face to hole edge (nominal 13 − 3.2 = 9.80; a new in-tolerance case can start at 12.9 − 3.215 = 9.685, so no absolute limit)	retire the end if any hole's axial length has grown, or its ligament has shrunk, by more than 0.05 from that hole's own baseline
R8	Hardness and conductivity baseline	op 160, after the first proof	HRB or Webster B, and eddy-current %IACS, at θ 50°, 130°, 230°, 310° (between the holes): aft end z 5 … 25 and z 100; fwd end z 665.8 … 685.8 and z 590.8	record, same instrument and scale for life (6061-T6: about 95 HB (about 60 HRB), Webster B ≥ 15, 40 … 46 %IACS)
R9	Proof OD, SRM100-001	before and after every proof	stations S1 … S5 at z 60, 172.7, 345.4, 518.1, 630.8 from the aft face, on removable tape; diameters through θ 45° and 135°, micrometer 0.001, tube temperature at the two readings within 2 °C	permanent set ≤ 0.02 at every station
R10	After every firing, before re-use	each firing	temperature labels (note 8), read and recorded before extraction; hardness and %IACS on the aft end at the R8 positions; holes as R7; seal zones visual (no scratch across the lay)	label ≥ 232 °C: re-proof or retire that end; > 46 %IACS, or hardness more than 10 % below its R8 baseline on the same instrument and scale: retire

Ra 3.2 (✓)

MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
6061-T6 drawn tube, ASTM B210M	2234.4 g	ISO 2768-mK	Ra 3.2	ISO 13715 -0.3 / +0.3

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
DETAIL DRAWING	NOT CHECKED - FOR REVIEW	1:2		mm

LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	MOTOR CASE Case tube ϕ100 × 4 × 690.8, bare 6061-T6, 2 × 18 radial bolt holes

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	-	2026-09-23	SRM100-101	A	1 / 1